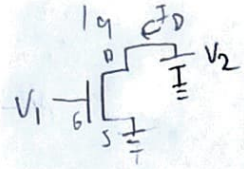
1

1a



$\frac{W}{L} = 20$

a. $V_1 = .2$, $V_2 = .1$ $V_{DS} = .1$

cutoff, $I_D = 0A$

b. cutoff, $I_D = 0A$ $V_{DS} = .6$

c. $V_{GT} = .5 - .39 = .11$ $V_{ds} = .2$

$.11 < .2$

Saturation, $I_D = 2.42 \times 10^{-5} A$

d. $V_{DS} = .8$

Saturation, $I_D = 2.42 \times 10^{-5} A$

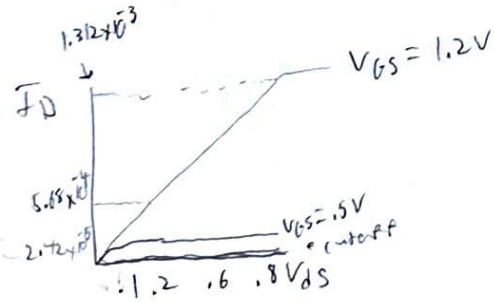
e. $V_{GT} = 1.2 - .39 = .81$

$.81 > .2$

linear, $I_D = 5.68 \times 10^{-4} A$

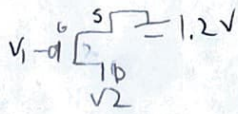
$V_{DS} = .8$

f. linear, $I_D = 1.312$ mA



1b

Dylan Thornton



a. $.2 + .27 - 1.2 = -.73$ $V_{GS} = -1V$

$.1 - 1.2 = -1.1$

Saturation, $I_D = 3.1924 \times 10^{-4} A$

b. $.6 - 1.2 = -.6$ $V_{GS} = -1V$

linear, $I_D = 3.096 \times 10^{-4} A$

c. $.5 - 1.2 + .27 = -.43$ $V_{GS} = -.7$

$V_{DS} = -1$

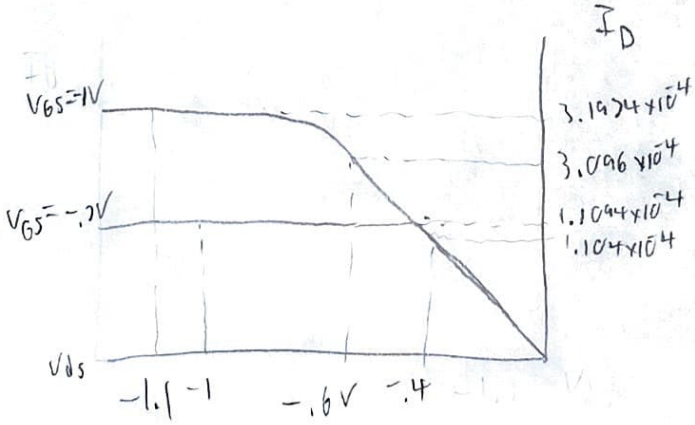
Saturation, $I_D = 1.1094 \times 10^{-4} A$

d. $.8 - 1.2 = -.4$ $V_{DS} = -.4$

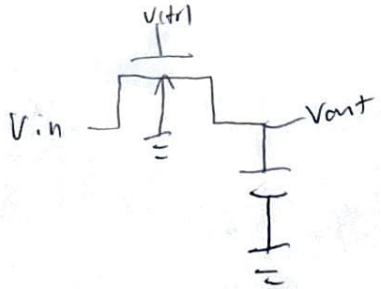
linear, $I_D = 1.104 \times 10^{-4} A$

e. cutoff, $I_D = 0A$

f. cutoff, $I_D = 0A$



2.



a. Vout = 0V

b. Vout = .4V

c. Vout = 0V

d. Vout = 1.2V (assuming V_T is < .3)

e. No required voltages are specified, so I will assume 0V and 1.2V is what we want. To do this, pulse between 0V and 1.5V